(Q No 2) part (B)

Start with sum = 1 (every num above 1 has 1 as a proper division) then i for 2 while $i \times i <= N$, when i divides n add i and n/i. if $i = n/i$ (a square root) add it only once. n itself is never added

i	divides n?	added	sum
start	-	1	1
2	yes	2 + 110 = 112	113
3	No	-	113
4	Yes	4 + 55 = 59	172
5	Yes	5 + 44 = 49	221
6	No	-	"
7	No	-	"
8	No	-	"
9	No	-	"
10	Yes	10 + 22 = 32	253
~11	Yes	11 + 20 = 31	284

the loop start at i = 14 (15×15 = 225 > 220) the sum is 284, the same result as the slow result version, found with 13 tests instead of 110. for 284 the loop runs i = 2 to 16 finds 2 & 142, 4 & 71 gives 1+142+4+71 = 220